KERNEL FORGE · UNIVERSITY OF YAOUNDÉ I

SELECTED · ORION HACKATHON — PHASE 2

UniFlow

The smart, modular, offline-first university platform — built for Africa.

Run your university, even when the network and the power go down.

PITCH DECK

Product presentation and value proposition

uniflow.kernelforge.codes

Version 3.0 · Complete redesign · August 2026

THE PROBLEM

A real constraint, not an assumption

Four public indicators, and two documented realities from the ground in Cameroon.

6 to 8 hrs / day

of scheduled power cuts, in many neighborhoods, since late 2024

Eneo, via Investir au Cameroun

41.9%

of the Cameroonian population connected to the internet

DataReportal / Kepios, 2026

47,952

students enrolled at UY1 in 2025–2026

UY1, key figures 2025–2026

3,000 students

in some lecture halls originally designed for 500 seats

Thot Cursus, 2022 ; Tepe et al., 2024

DOCUMENTED CASE · MANUAL MANAGEMENT

In several African universities, administration remains entirely manual: a former student can spend years retracing their own academic record — paper files, duplicates, no traceability.

Testimony collected by ACP, 2026

DOCUMENTED CASE · ENERGY REALITY

In Cameroonian university towns, power cuts last an average of 222 hours per month (up to 250 hrs during peaks). Every evening, students head to the lecture halls to make use of the generator.

Djoufack, K.-O. et al., 2022 (Univ. of Dschang)

THE SOLUTION

UniFlow: a platform designed for the field, from day one

A general-purpose, modular, open-source (MIT) university management foundation — not retrofitted from a Western product. Our ambition goes further: not just a digital campus, but a smart, interactive campus, powered by AI and connected IoT modules like Vigie.

THE KEY DIFFERENTIATOR

Offline-first

Deferred sync via a local queue (Outbox pattern): UniFlow keeps working through network or power outages — a daily constraint no direct competitor natively handles.

Enrollment management

Enrollment by course unit, program, and level.

Smart timetables

Automatic generation and self-adjustment.

QR-code attendance

Fast, reliable check-in, usable in real time.

Communication & video

Real-time notifications, classes in-person or online.

Assessment & analytics

Grades, dashboards, performance tracking.

THE MARKET

Considerable growth potential

From the pilot university to all of Sub-Saharan Africa.

TAM

≈ 9 million

higher-education students in Sub-Saharan Africa — a gross enrollment rate of just 9%, versus 43% worldwide: a largely untapped potential.

World Bank / UNESCO Institute for Statistics

SAM

≈ 700,000

higher-education students in Cameroon — 11 state universities and 476 private institutions (IPES).

MINESUP, via Africa24, 2023

SOM

47,952

students at UY1 — our launch pilot.

UY1, key figures 2025–2026

TAM · Total addressable market — SAM · Serviceable market (Cameroon) — SOM · Obtainable market (UY1 pilot)

THE PRODUCT

UniFlow Sentinelle: the campus, down to the physical layer

This is where UniFlow becomes a smart, interactive campus: deliberately low-cost, rugged hardware (Raspberry Pi rather than dedicated servers) extends the platform to AI-driven campus safety and health.

Health Module

Pre-diagnosis kiosk at the infirmary: connected medical sensors (pulse oximeter, blood-pressure monitor, thermometer) and a triage AI that classifies urgency — green, orange, red.

Vigie (Watch) Module

AI-based fall and anomaly detection at the edge, functional even offline, at sensitive points across campus.

Cross-module synergy

A fall detected by Vigie automatically triggers Health's emergency protocol, without anyone needing to reach a kiosk.

Flutter
Mobile · Desktop · Web

NestJS + Prisma
Modular API

PostgreSQL
Supabase

Raspberry Pi
Edge IoT

VALIDATION & TRACTION

What we've already proven

Execution already underway — not just an idea on paper.

Shortlisted — Orion Hackathon, Global 2026

Phase 2 of evaluation, a competition run with Google Developer Groups (GDG) and UnsaidTalks.

Complete technical specification

Finalized specification for the UniFlow academic module.

Team of 8 developers already in place

Actively engaged in development, not just on paper.

Live demo available

uniflow.kernelforge.codes

NEXT VALIDATION STEP

Real-world pilot at UY1

Testing the academic MVP with real students and teachers, before extending to other Cameroonian universities.

V1	Months 1–6 — JWT+RBAC, academics, attendance, offline sync
V2	Months 7–9 — Exams, library, Sentinelle pilot
V3	Months 10–15 — Predictive AI, regional expansion

TEAM & GOVERNANCE

Our talent

Meet the team and their areas of expertise.

8 Members total · 4 Frontend Engineers · 3 Backend & DB Engineers · 1 Lead & Architecture

LEAD ARCHITECT

NF

Nghomsi Feukouo Ravel

Architecture & Direction

@Archlord12345

WEB DESKTOP

AR

Aliyatou Rachid O. T.

Frontend Desktop & Web

@aliya-nadi

MOBILE APP

MJ

Mandeng Judith O.

Frontend Mobile App

@oceannemj

BACKEND & DB

MW

Meli William

Backend APIs & DB

@WilliamMeli-27

MOBILE APP

FM

Febnchak M. B. Sandra

Frontend Mobile App

@FEBNCHAK

NESTJS BACKEND

HY

Hassane Youssouf O.

Backend Microservices

@hawadja1

DATABASE ARCHITECT

MA

Mokam Ange

DBMS & Infrastructure

@Ange55-star

FULLSTACK UI

SK

Sineng Kengni Juvenal

Cross-platform

@skjuv

COMPETITION

Built for the field, not retrofitted

Existing academic ERPs (SIGES, Moodle, Western solutions) assume a stable network and hard-currency budgets — two conditions rarely met on the ground in Cameroon.

EXISTING SOLUTIONS

Designed for a context where network and power are a given.

UNIFLOW

Designed offline-first and low-cost, from the ground up.

No offline mode

Connection required

Most academic ERPs stop working at the first network or power outage — a daily constraint in Cameroon.

Hard-currency licensing

Open source (MIT)

100 FCFA / month / user

International ERPs bill in dollars or euros, out of budget for most African universities. UniFlow bills in FCFA.

FINANCIAL PROJECTIONS

A trajectory built on a low-cost model

Minimal development and infrastructure cost, for an adoption trajectory directly aligned with the market defined earlier.

1,510 – 2,040 hrs

estimated MVP development

37 – 126 $

infrastructure cost / month in production

100 FCFA

entry price / month / individual user

GROWTH SCENARIO — TARGET ACTIVE USERS (ESTIMATE)



THE ASK

What UniFlow needs to scale

Seed funding

Cover production infrastructure and MVP development time through to the UY1 pilot.

Institutional partnership

A pilot partnership with the University of Yaoundé I to deploy UniFlow under real conditions starting next semester.

Mentorship & guidance

Guidance to structure the project's next steps beyond the hackathon.

ESTIMATED SEED BUDGET

1.5M – 8M FCFA

Estimate based on our own development costs (1,510–2,040 hrs) at Cameroonian market hourly rates — to cover development through the UY1 pilot.

APPENDIX · BUSINESS MODEL

Progressive monetization, designed for adoption

Free access during the launch phase, then a switch to a subscription model starting in September.

CURRENT PHASE — FREE

Until end of September: open access to drive adoption.

FROM SEPTEMBER — SUBSCRIPTION

Switch to a paid monthly subscription model.

Independent accounts

100 FCFA

For individual users not affiliated with a partner institution — full platform access, simple direct billing.

University accounts

Subscription

Rate set by agreement, based on user volume

The university takes out a blanket subscription and covers access for all its students and staff — no individual billing.

APPENDIX · PARTNERSHIPS

An ecosystem already engaged, a clear trajectory

From the hackathon ecosystem to deployment across Cameroonian universities.

CURRENT ECOSYSTEM — ORION HACKATHON GLOBAL 2026

- Google Developer Groups (GDG)
- UnsaidTalks
- INFO

TARGET PARTNERSHIP

University of Yaoundé I

Pilot institution for the V1 real-world deployment — 47,952 students, the same university all KERNEL FORGE team members come from.

PLANNED EXTENSION

V1 · Months 1–6
UY1 pilot
Academics, attendance, offline sync

V2 · Months 7–9
State universities
Sentinelle pilot, public API

V3 · Months 10–15
Regional expansion
Predictive AI, other countries

UniFlow



A tool designed to work locally first, and sync later.

UniFlow centralizes a university's academic, administrative, and educational management — open source (MIT) and resilient to power and network outages. Tested first at the University of Yaoundé I, built for universities across Africa.

LIVE DEMONSTRATION

uniflow.kernelforge.codes

The platform will be demonstrated live, at this address, during the presentation.

Thank you for your attention!

KERNEL FORGE · University of Yaoundé I